The Quadratics Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: FACTOR, ROOTS, VERTEX and MODEL (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (FACTOR)
- ☐ **A2** (ROOTS)
- ☐ **A3** (VERTEX)
- ☐ **A4** (ROOTS)
- ☐ **A5** (FACTOR)
- ☐ **A6** (MODEL)

Entrées

10 pts each · choose 4

- ☐ **E1** (FACTOR)
- ☐ **E2** (FACTOR)
- ☐ **E3** (ROOTS)
- ☐ **E4** (ROOTS)
- ☐ **E5** (VERTEX)
- ☐ **E6** (MODEL)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (MODEL)
- ☐ **S2** (ROOTS)
- ☐ **S3** (VERTEX)
- ☐ **S4** (MODEL)

My skill check (circle each skill you have covered): FACTOR · ROOTS · VERTEX · MODEL

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 FACTOR ☐ My pick · 5 pts

Solve by factoring: $x^2 + 7x + 12 = 0$

A2 ROOTS ☐ My pick · 5 pts

Solve using square roots: $2x^2 - 50 = 0$

A3 VERTEX ☐ My pick · 5 pts

Find the axis of symmetry of the parabola $y = x^2 - 8x + 3$.

A4 ROOTS ☐ My pick · 5 pts

Use the discriminant to decide how many real solutions $x^2 + 4x + 7 = 0$ has. Show the value you find.

A5 FACTOR ☐ My pick · 5 pts

Solve by factoring: $3x^2 - 12x = 0$

A6 MODEL ☐ My pick · 5 pts

A rectangle is 4 m longer than it is wide, and its area is 45 m². Write an equation for the width w, solve it, and give both dimensions.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 FACTOR ☐ My pick · 10 pts

Solve by factoring: $3x^2 - 10x - 8 = 0$

E2 FACTOR ☐ My pick · 10 pts

Ana solved $(x - 2)(x + 3) = 14$ by writing $x - 2 = 14$ or $x + 3 = 14$, so $x = 16$ or $x = 11$. Explain her mistake, then solve the equation correctly.

E3 ROOTS ☐ My pick · 10 pts

Solve by completing the square: $x^2 + 8x + 5 = 0$. Give exact answers.

E4 ROOTS ☐ My pick · 10 pts

Solve using the quadratic formula: $2x^2 - 3x - 4 = 0$. Give exact answers.

E5 VERTEX ☐ My pick · 10 pts

For $y = -2x^2 + 8x - 3$, find the axis of symmetry and the vertex. Is the vertex a maximum or a minimum? Explain.

E6 MODEL ☐ My pick · 10 pts

A rectangle has perimeter 40 m and width w. Write its area A in terms of w. What width gives the greatest area, and what is that area?

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 MODEL ☐ My pick · 20 pts

A rocket leaves a 24 ft platform. Its height is $h = -16t^2 + 40t + 24$ (feet, t in seconds). **(a)** How high is it at launch? **(b)** When does it hit the ground? Show how you decide which solution to use. **(c)** Find its maximum height and when it happens.

S2 ROOTS ☐ My pick · 20 pts

Consider $x^2 + kx + 9 = 0$. **(a)** Find every value of k that gives exactly one real solution. **(b)** For the positive k from (a), solve the equation. **(c)** Explain, using the discriminant, why k = 4 gives no real solutions.

S3 VERTEX ☐ My pick · 20 pts

A parabola has vertex (3, −4) and passes through the point (1, 4). **(a)** Use $y = a(x - 3)^2 - 4$ to find a. **(b)** Write the equation in standard form. **(c)** Find the x-intercepts exactly.

S4 MODEL ☐ My pick · 20 pts

A 20 ft by 12 ft garden has a walkway of equal width x all the way around it. Garden and walkway together cover 660 ft². Sam wrote $(20 + x)(12 + x) = 660$. **(a)** Explain his error. **(b)** Write and solve the correct equation. **(c)** State the walkway width and the outer dimensions.